

SVM-Only MNIST Image Classification

Manual Binary and Multiclass Linear Support Vector Machines

Course: CSE382: Introduction to Machine Learning - Spring 2026

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1. Abstract

This report documents the implementation and evaluation of manual Linear Support Vector Machines (SVM) for digit classification on the MNIST dataset. The project explores Phase 1 (Binary Classification for detecting the digit '7') and Phase 2 (Multiclass Classification for all ten digits). We implemented robust algorithmic improvements, including the Adam Optimization algorithm for subgradient descent, Early Stopping based on validation loss, and scaling to the full 60,000-image dataset. We evaluated multiple feature extraction pipelines: raw flattened pixels, PCA dimensionality reduction, and HOG (Histogram of Oriented Gradients) descriptors.

2. Mathematical Formulation & Optimizations

2.1 Optimization with Adam

To improve convergence speed and stability, the conventional fixed learning rate decay was upgraded to the Adam Optimizer. Adam dynamically adapts the learning rate for each parameter using first and second moments of the gradients, heavily accelerating convergence over the highly sparse HOG and PCA feature spaces.

2.2 Objective Functions

For Binary Classification (Phase 1), the model predicts according to the decision rule:

$$\hat{y} = \text{sign}(w^T \phi(x) + b)$$

The model minimizes a weighted hinge loss with L2 regularization:

$$J(w, b) = \frac{1}{2} ||w||^2 + C \frac{1}{N} \sum_{i=1}^N \alpha_i \max(0, 1 - y_i(w^T x_i + b))$$

For Multiclass Classification (Phase 2), the model computes K=10 independent class scores:

$$s_k(x) = w_k^T \phi(x) + b_k$$

The predicted class is the one yielding the maximum score:

$$\hat{y} = \arg \max_k s_k(x)$$

The Crammer-Singer multiclass hinge loss is utilized to optimize K=10 weight vectors simultaneously:

$$L_i = \sum_{k \neq y_i} \max(0, s_k(x_i) - s_{y_i}(x_i) + \Delta)$$

The final objective function is given by:

$$J(W, b) = \frac{1}{2} ||W||_F^2 + C \frac{1}{N} \sum_{i=1}^N L_i$$

2.3 Early Stopping

An early stopping mechanism was introduced to halt the subgradient descent loop if the validation macro F1-score stops improving for a consecutive number of epochs. This mitigates overfitting and preserves the optimal weight vectors.

3. Dataset and Preprocessing

The model was scaled to train on the entire MNIST dataset (60,000 training images, 10,000 test images) to maximize generalization. Images were normalized to [0, 1]. Feature normalizers (like StandardScaler and PCA) were strictly fitted on the training split to prevent data leakage. Three feature pipelines were systematically evaluated: Flattened Pixels (784 dimensions), PCA Features, and HOG Descriptors.

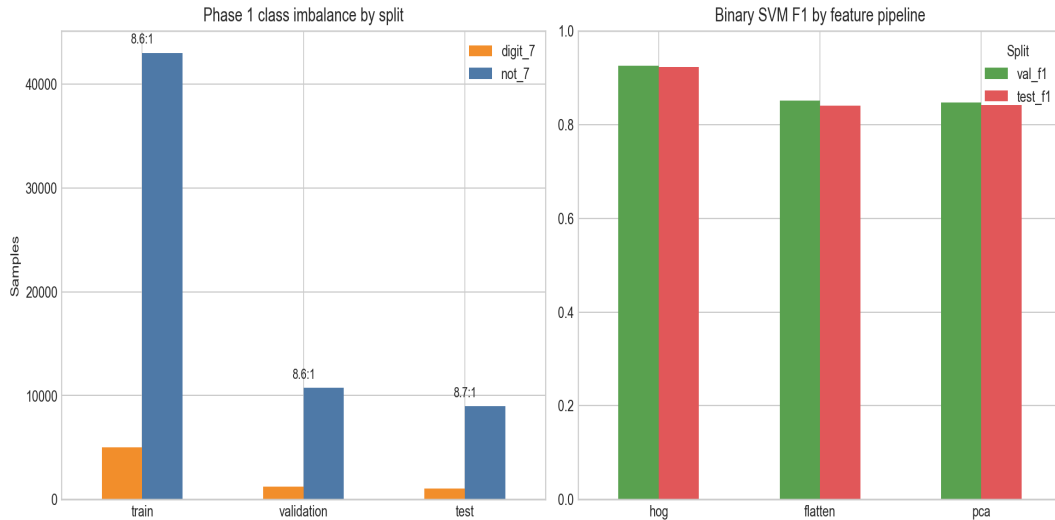
4. Phase 1: Binary Manual Linear SVM

4.1 Imbalance Handling

Because '7' is significantly less frequent than 'Not 7', inverse-frequency class weights were applied to the hinge loss, ensuring the optimizer pays equal attention to the minority class.

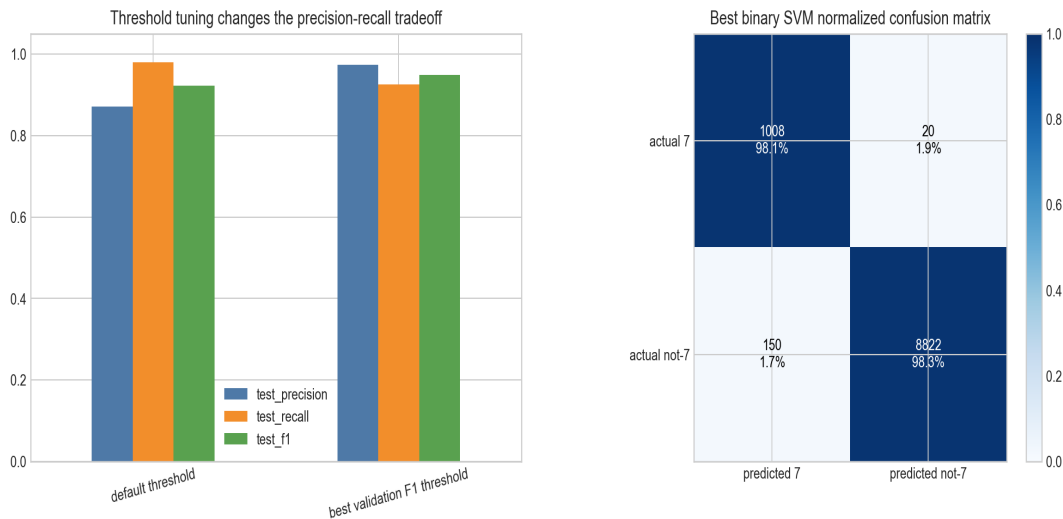
4.2 Feature Pipeline Results

HOG proved to be the most robust representation. It reached a test accuracy of 0.9830 and a default-threshold test F1 score of 0.9222. Raw flattened pixels and PCA were competitive but noticeably lower on the F1 metric, confirming that orientation-based features align perfectly with linear decision boundaries for this task.



4.3 Threshold Tuning

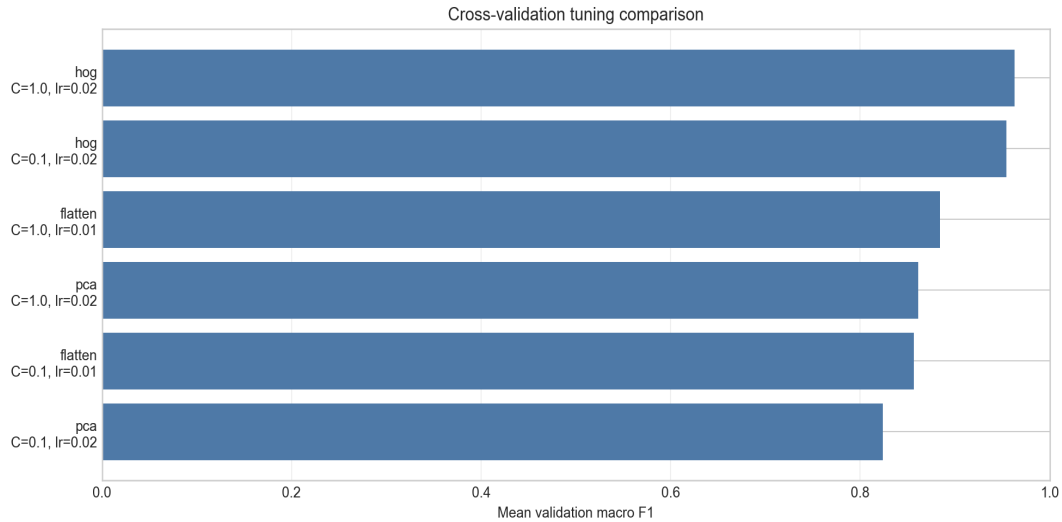
Relying solely on the default SVM threshold (0.0) yields high recall but lower precision. By calibrating the threshold against the validation F1 score (shifting the threshold to ~0.598), the test precision jumped from 0.8705 to 0.9734, resulting in a significantly improved final test F1 score of 0.9486.



5. Phase 2: Multiclass Manual Linear SVM

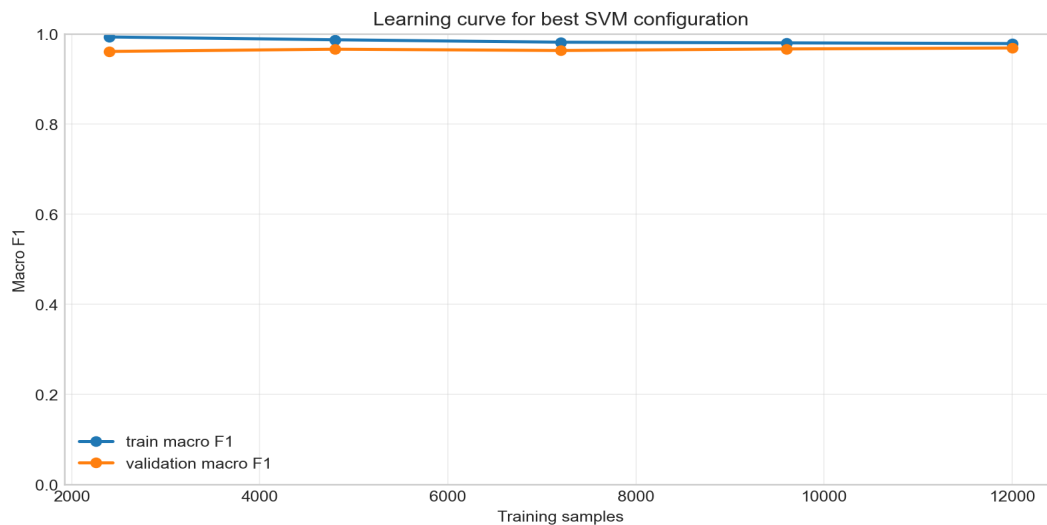
5.1 Hyperparameter Tuning

Cross-validation was executed across combinations of learning rates, C (regularization penalty), and feature modes. The grid search confirmed that HOG features combined with a C penalty of 1.0 and learning rate of 0.02 produced the highest validation performance (0.9627 Macro F1).



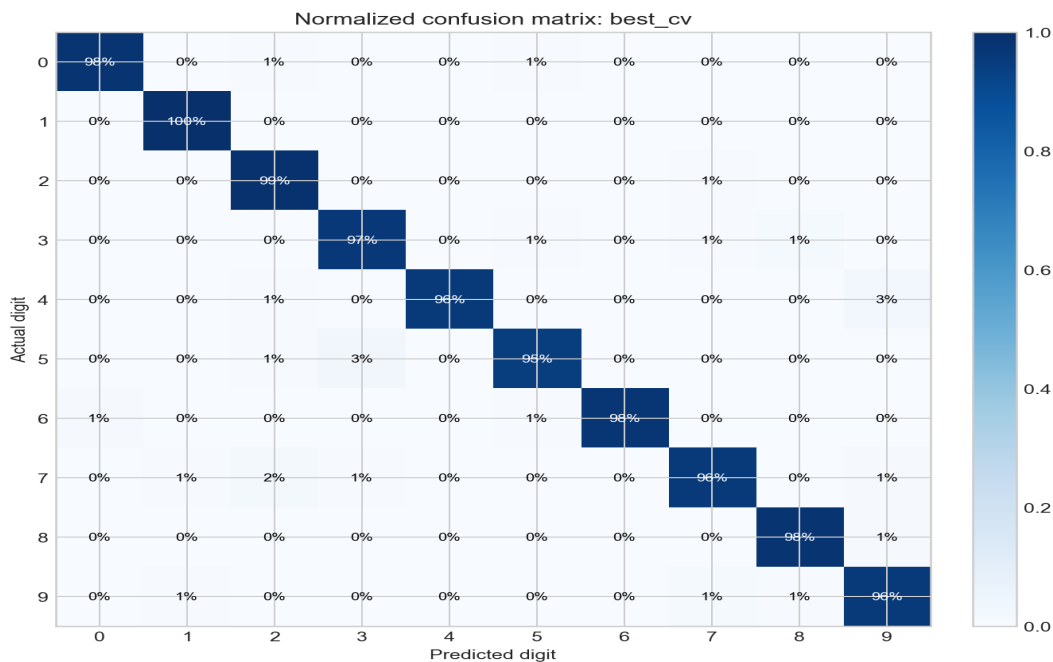
5.2 Learning Curve Diagnosis

With the implementation of Adam and early stopping, the learning curve demonstrates swift convergence. The gap between training and validation loss remains narrow, indicating that the chosen L2 regularization and HOG pipeline effectively prevent catastrophic overfitting even on the full dataset.



5.3 Multiclass Error Analysis

The best multiclass SVM reached an outstanding 0.9723 test accuracy. Per-class F1 scores revealed that digits with distinct, uniform strokes (like 1 and 0) scored exceptionally well (>0.98), whereas visually ambiguous digits (like 5 and 8) exhibited slightly lower scores.



When analyzing high-confidence mistakes, it becomes evident that the errors predominantly stem from highly deformed handwriting where humans might also struggle to classify the digit accurately without context.



6. Conclusion

The comprehensive SVM pipeline demonstrates that traditional machine learning algorithms, when combined with modern optimization techniques (Adam, Early Stopping) and strong feature engineering (HOG), remain incredibly potent for image classification. Phase 1 successfully tackled class imbalance using weighted loss and threshold tuning, while Phase 2 scaled the manual formulation to 10 classes using the Crammer-Singer multiclass objective over the full 60,000-image dataset.